Rogue Fin Design: Summary Report

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Repository: `https://github.com/sidd-somadder/rogue-fin-design`

This report summarizes the fin redesign for NYU Rogue Aerospace's 2026–27 FAR-OUT rocket; the full record of each decision is in the working design logbook. The goal is to reduce fin planform area, and with it fin drag, while meeting the competition's stability and damping requirements and the team's flutter margin. Three design iterations are complete. The current fin is sized for the full-scale rocket, which reaches $M \approx 0.46$ in simulation, and will first fly on the November 8 sub-scale launch; the four sub-scale fins are fabricated and have been delivered for carbon-fiber layup.

My Role

I am the Flight Dynamics engineer responsible for the fin redesign. My own contributions:

- **Requirements.** Set how the FAR-OUT stability and damping rules are checked: the worst case of 30 turbulent OpenRocket runs in a 20 mph crosswind, stricter than the rule.
- **Analysis tools.** Wrote the five Python tools in the repository: `damping.py` (damping over a full flight, Mandell's method with jet damping and compressibility), `drag_estimate.py` (strip-summed skin-friction drag), `flutter.py` (NACA TN 4197), `profile_gen.py` (section coordinates), and `xfoil_wrapper.py` (batch XFOIL runs). The damping and drag methods are written up in *Rocket Damping* and *Preliminary Fin Drag Study*.
- **Verification.** Traced a fourfold error in the damping ratio to an OpenRocket listener that measured moment arms from the nose tip instead of the CG; with Mandell's aspect ratio corrected, my tool matches OpenRocket's normal-force slope within 1%.
- **Trade study.** Selected the Iteration 3 fin: 16.7% less area and an estimated 11.8% less fin drag than the previous full-scale fins, with a 1.66 cal worst-case rail-exit margin and a flutter margin of 3.5.
- **Fabrication.** Cut, sanded and chamfered the four sub-scale fins for carbon-fiber layup (Figure 2).

Team context: the 45° edge chamfer is the team's existing cross-section, the decision to size the fins to full scale came from team leadership, and the Aerostructures team runs the root FEA.

Requirements

Requirement	Value	Source	Evaluated at
Static stability	≥ 1.5 cal, rail exit to burnout	FAR-OUT 5.1.5	20 mph wind, 15% turbulence, worst case of 30 runs (stricter than the rule, which gives no wind speed)
Damping ratio	$0.03 \leq \zeta \leq 0.5$ (0.05–0.3 recommended)	FAR-OUT 5.1.5	30 mph wind; rail exit and coast minimum
Flutter	$V_f \geq 2\,V_{max}$	Team	NACA TN 4197 with conservative inputs
Root chord	1–2 cal (5–10 in)	Aerostructures	Bonding area and root bending

All flights are simulated in OpenRocket with a 600 in rail at 40 °F and the launch-site altitude of 410 ft.

Approach

1. **Size for rail exit.** Rail exit in a crosswind, not nominal flight, sets the fin size, so candidates are sized to the smallest area whose worst-case rail-exit stability over repeated turbulent simulations meets 1.5 cal.
2. **Check flutter and damping.** Flutter speed from NACA TN 4197; damping ratio from `damping.py`, built on Mandell's equations with a compressibility correction.
3. **Trade study.** Passing candidates are compared on estimated skin-friction drag, flutter margin, and manufacturability. Simulated apogee was dropped as a metric because its run-to-run scatter (about 17 ft) exceeded the differences between candidates.

4. **Cross-section study.** Edge treatments are compared by XFOIL viscous analysis where it converges and by 2D CFD in ANSYS Fluent where it does not, weighed against manufacturing effort and repeatability.
5. **Freeze & Verify.** Hand design to Aerostructures for FEA and also verify with 3D CFD. Reach a consensus between teams on where the part fails and needs a redesign, if any.
6. **Manufacture and Launch.** Wood cores are cut and sanded (complete for the sub-scale fins), then laid up in carbon fiber. Validate performance against sub-scale flight data.

Iteration History

Iteration	Selected planform (in)	Area (in^2)	Outcome and reason for closing
1 (9/30–10/6)	7×3×4	20.0	Sized to a 15% area-reduction target; drag −12.0%, flutter margin 4.4. Closed: an area target does not find the minimum fin, and stability was in percent, not the rules' calibers.
2 (10/7–10/15)	5.5×3×4.25	18.06	Re-sized to the rail-exit requirement at 30 mph: 1.58 cal minimum over 20 runs (baseline 1.29, failing), area −24.8%, drag −16.4%, flutter margin 4.6; 7° double wedge. Closed when leadership asked for fins designed to full scale.
3 (10/15–)	**7.5×2.5×5**	**25.0**	Sized against the full-scale 10×2×5 fins (30.0 in^2): rail-exit minimum 1.66 cal, area −16.7%, drag −11.8%, flutter margin 3.5 vs. 3.1, coast $\zeta \approx 0.06$. 45° chamfer for the sub-scale.

Planforms are root × tip × height; Iterations 1–2 compare to last year's sub-scale fins (9×3×4, 24 in^2).

Current Design (Iteration 3)

A 7.5 in root × 2.5 in tip × 5.0 in height cropped delta with a 0.25 in wood core and tip-to-tip carbon fiber, keeping the team's 45° chamfered edges because late material deliveries left no time for a new cross-section. Among the passing candidates, it was chosen over a 7.5 in/3 in tip (equal stability within 0.02 cal, more drag) and over 2 in tips (too little material for a longer edge treatment later). A 6×2×5.5 fin with better stability and drag was rejected for its short root and long span, which reduce bonding area and raise root bending stress.

DESIGN SELECTION: (1) 7.5 inch root × 2.5 inch tip × 5 inch height cropped delta planform; (2) 45° chamfered edges (the team's existing cross-section).

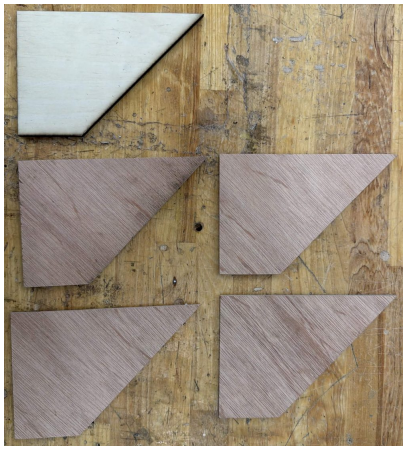


(a) Top view (b) Auxiliary view

Figure 1: Iteration 3 fin geometry: 7.5×2.5×5.0 in planform with 45° chamfered edges.

Fabrication

I fabricated the four sub-scale fins from marine-grade wood stock. Each fin was rough-cut on the bandsaw and brought to the planform on the belt sander. The 45° edge chamfers were then sanded with a hand-sanding jig so the edges match across all four fins, followed by corrections by hand. The finished cores were delivered for carbon-fiber layup ahead of the November 8 launch.



(a) Four fins, birch reference fin at top left



(b) Tip chamfer



(c) LE/TE chamfers

Figure 2: Sub-scale fins from marine-grade wood, before carbon-fiber layup. The edges were sanded with a jig so all four fins share the same chamfer geometry.

Key Findings/Lessons

- **Read the rules first.** Iteration 1 was sized to an area goal before the rules were read in detail; the rail-exit requirement in calibers changed the design driver entirely.
- **Fin height drives stability; chord drives drag less than area suggests.** Estimated drag falls about 0.7 times as fast as area, because shorter chords raise both the friction coefficient and the thickness ratio. Height is the most effective stability lever but raises root bending.
- **Simulation tools need independent checks.** An OpenRocket damping listener measured moment arms from the nose tip instead of the CG, overstating ζ about fourfold. The independent tool, with Mandell's aspect ratio corrected to the fin plus its mirror image, matches OpenRocket's normal-force slope within 1%.
- **XFOIL fails on sharp edges.** Its viscous solver does not converge on chamfered sections, so those comparisons move to CFD, validated first against XFOIL on the NACA 0005.
- **Manufacturing limits the cross-section.** Knife-edge leading and trailing edges snapped during sanding, so edges must keep a small flat or a rounded leading edge.

Next Steps

- **Sub-scale flight (Nov. 8):** compare measured apogee and stability with OpenRocket predictions with Avionics.
- **Iteration 4 cross-section:** attempt prototyping a rounded leading edge and beveled trailing edge with ShopBot machining and a jig; run drag comparison in validated 2D CFD.
- **Full-scale checks:** rerun flutter at the maximum-velocity altitude with the carbon-fiber skins, confirm the static margin through burnout as the hybrid's CG shifts, and complete the root FEA. Possibly develop more precise flutter estimate technique utilizing ANSYS Modal Analysis.